Mono-Objective Optimization and Its Usage for Identifying Model Parameters of Semiconductor Elements of Radio Electronics

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Implementation of Analog Systems



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A Modified Method
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Conclusion

1. Outline

- Definition of criterion for a minimization algorithm
- Modification of the Levenberg-Marquardt procedure
- Normalization of system of equations
  - Normalization of differences
  - Normalization of Jacobian (a new feature)



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1. Outline

- Definition of criterion for a minimization algorithm
- Modification of the Levenberg-Marquardt procedure
- Normalization of system of equations
  - Normalization of differences
  - Normalization of Jacobian (a new feature)
- Practical results of measurements and identifications of model parameters
  - Two criteria of successfulness of the identification
  - BJT KC508 (a Czech equivalent of BC108)
  - Microwave BJT KT391
  - Enhancement-mode PMOSFET 2N3608
  - Enhancement-mode power NMOSFET BUZ345
  - Depletion-mode NMOSFET KF521
  - PJFET 2N2498



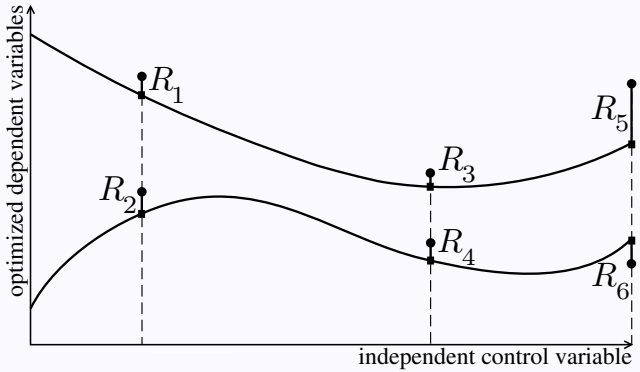
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2. Minimization of the Sum of Squares

The algorithm seeks to minimize the sum of squares of differences between measured data and the outcome of an analytic model

$$S\left(x_1,\ldots,x_n\right)=\sum_{k=1}^{m}R_k^2\left(x_1,\ldots,x_n\right),\ n\leqq m,$$

where the unknown optimized parameters are marked by $x_1,\ldots,x_n$, and $R_k,\ k=1,\ldots,m$ are the differences.



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3. A Modified Method

The standard Levenberg (1st) and Levenberg-Marquardt (2nd) procedures

$$\left(\boldsymbol{J}^{\mathrm{t}}\boldsymbol{J}+\lambda^{(l)}\mathbf{1}\right) \qquad \Delta\boldsymbol{x}^{(l)} = -\boldsymbol{J}^{\mathrm{t}}\boldsymbol{r}, \;\; \boldsymbol{x}^{(l+1)} = \boldsymbol{x}^{(l)} + \Delta\boldsymbol{x}^{(l)}, \quad l = 1, \ldots, l_{\max}$$

$$\left(\boldsymbol{J}^{\mathrm{t}}\boldsymbol{J}+\lambda^{(l)}\mathrm{diag}\left(\boldsymbol{J}^{\mathrm{t}}\boldsymbol{J}\right)\right) \quad \Delta\boldsymbol{x}^{(l)} = -\boldsymbol{J}^{\mathrm{t}}\boldsymbol{r}, \;\; \boldsymbol{x}^{(l+1)} = \boldsymbol{x}^{(l)} + \Delta\boldsymbol{x}^{(l)}, \quad l = 1, \ldots, l_{\max}$$



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are completed using a natural criterion for selecting the scalar factor v (5, e.g.)

$$\lambda^{(1)} = 1,$$
$$\lambda^{(l+1)} = \frac{\lambda^{(l)}}{v}$$

at each iteration with a natural control of convergence

$$\underline{\texttt{if}}\ l > 1 \ \wedge\ S^{(l)} \geqq \min_{j=1,\ldots,l-1} S^{(j)}\ \underline{\texttt{then}}$$
$$\boldsymbol{x}^{(l+1)} := \boldsymbol{x}^{(l)},$$
$$\lambda^{(l+1)} := \lambda^{(l)}\, v^2,$$

see more at https://en.wikipedia.org/wiki/Levenberg-Marquardt_algorithm.



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4. Normalization of System of Equations

- Normalization of the differences

$$R'_k\left[\boldsymbol{x}^{(l)}\right] \triangleq w_k \frac{y_k^{(\text{ident})}\left[\boldsymbol{x}^{(l)}\right] - y_k^{(\text{meas})}}{\left|y_k^{(\text{meas})}\right| + y_k^{(\text{null})}}, \quad k = 1, \ldots, m$$

(The parameters $y_k^{(\text{null})}$ stabilize the differences when some measured values are near or equal to zero.)



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- Normalization of the Jacobian (this is an update of the method)

$$\frac{\partial R'_k\left[\boldsymbol{x}^{(l)}\right]}{\partial x_i} := w_k \frac{\partial y_k^{(\mathrm{ident})}\left[\boldsymbol{x}^{(l)}\right]}{\partial x_i} \frac{x_i^{(\max)} - x_i^{(\min)}}{\left|y_k^{(\mathrm{meas})}\right| + y_k^{(\mathrm{null})}},$$
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(The red expression is a result of the sensitivity analysis.)



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The above equation represents an assignment. Therefore, the solution of the linear system must be modified by another assignment

$$\Delta x_i^{(l)} := \Delta x_i^{(l)}\left[x_i^{(\max)} - x_i^{(\min)}\right], \quad i = 1, \ldots, n.$$



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5. Practical Results of Measurements and Identifications of Model Parameters of Electronic Elements

The root mean square and maximum deviations computed for all the following results are defined naturally

$$\mathrm{rms} = \sqrt{\frac{\sum_{i=1}^{n_p} \left(\frac{y_i^{(\mathrm{ident})} - y_i^{(\mathrm{meas})}}{y_i^{(\mathrm{meas})}} \right)^2}{n_p}} \times 100 \ \%,$$

$$\delta_{\max} = \max_{i=1,\ldots,n_p} \left| \frac{y_i^{(\mathrm{ident})} - y_i^{(\mathrm{meas})}}{y_i^{(\mathrm{meas})}} \right| \times 100 \ \%$$



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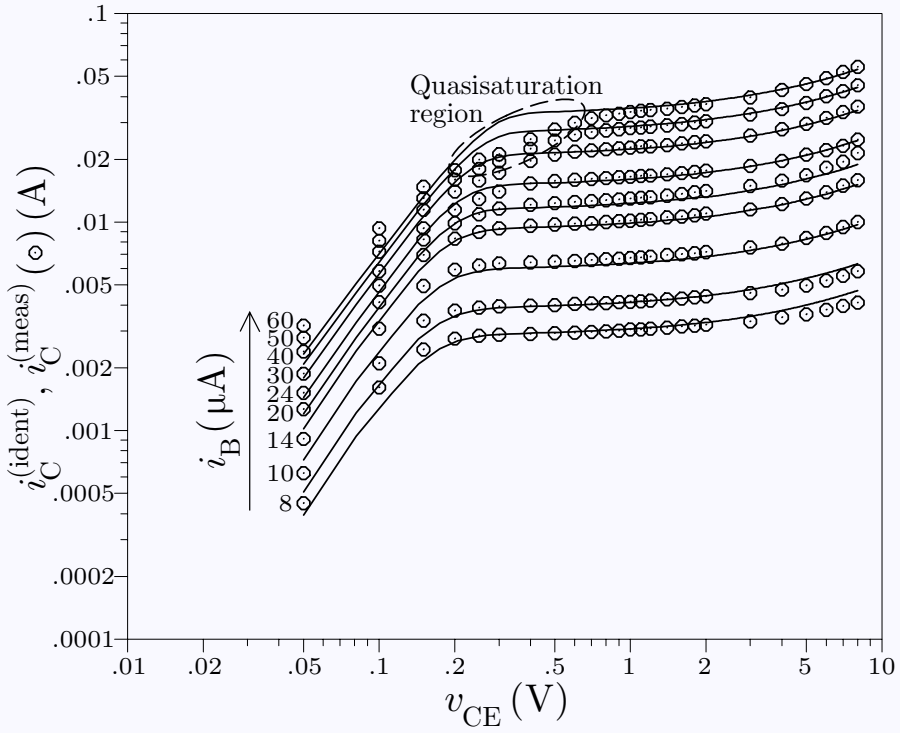
$$\delta_{\max} = \max_{i=1,\ldots,n_p} \left| \frac{y_i^{(\mathrm{ident})} - y_i^{(\mathrm{meas})}}{y_i^{(\mathrm{meas})}} \right| \times 100 \ \%,$$

where $y_i^{(\mathrm{ident})}$ and $y_i^{(\mathrm{meas})}$ are the identified and measured values, respectively, and n_p is the number of all measured points.



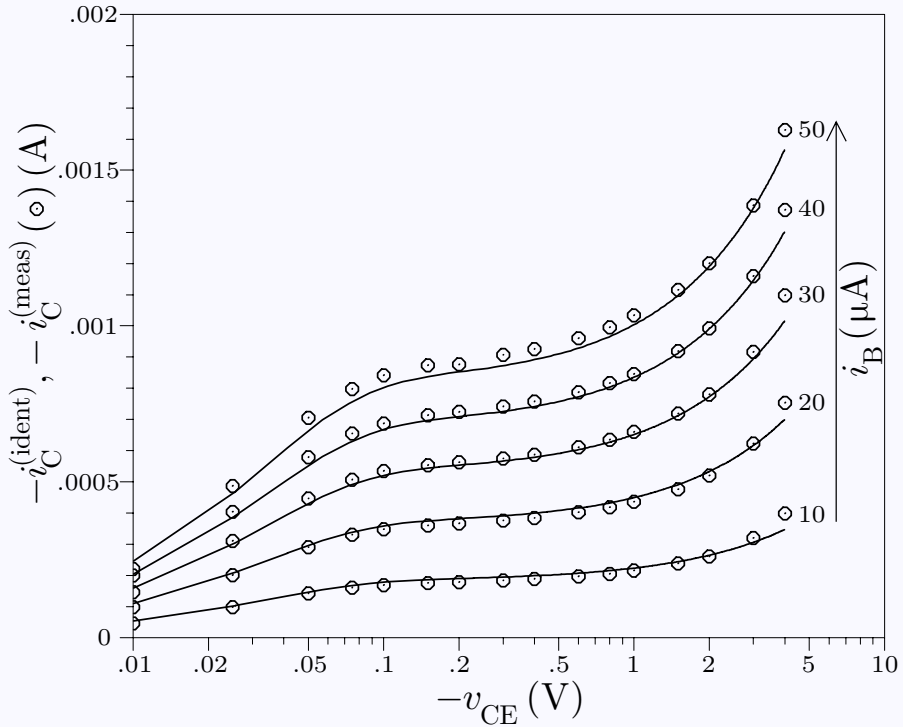
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Forward DC characteristics of BJT KC508 (this is a Czech equivalent of BC108, rms = 9.61 %, δ_{max} = 43.1 % – without a quasisaturation model part):



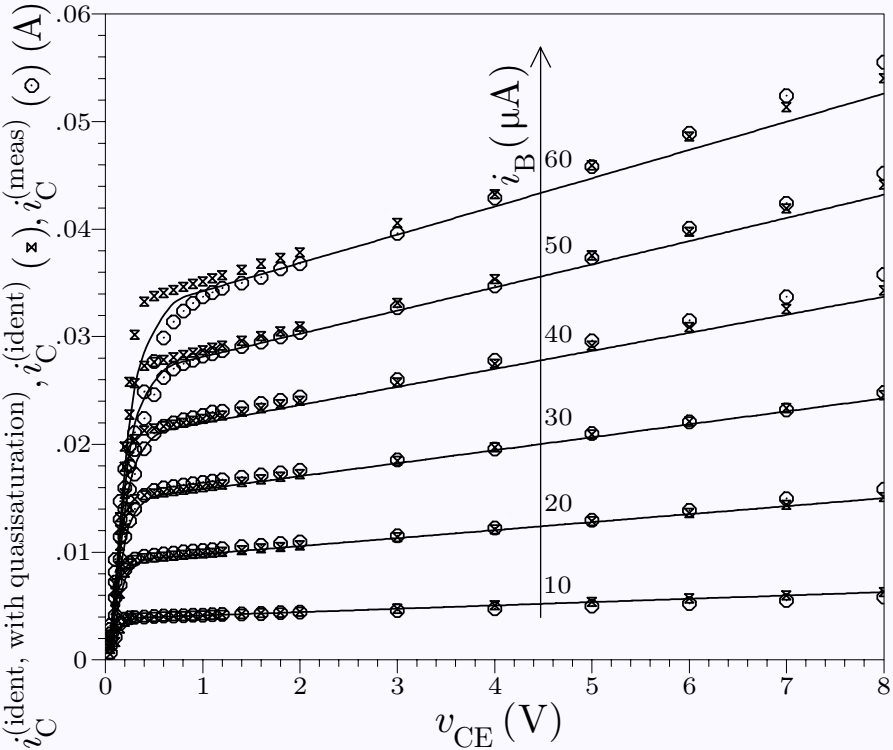
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Reverse DC characteristics of BJT KC508 (rms = 4.85 %, $\delta_{\max}$ = 20.0 % – again, without the quasisaturation model part):



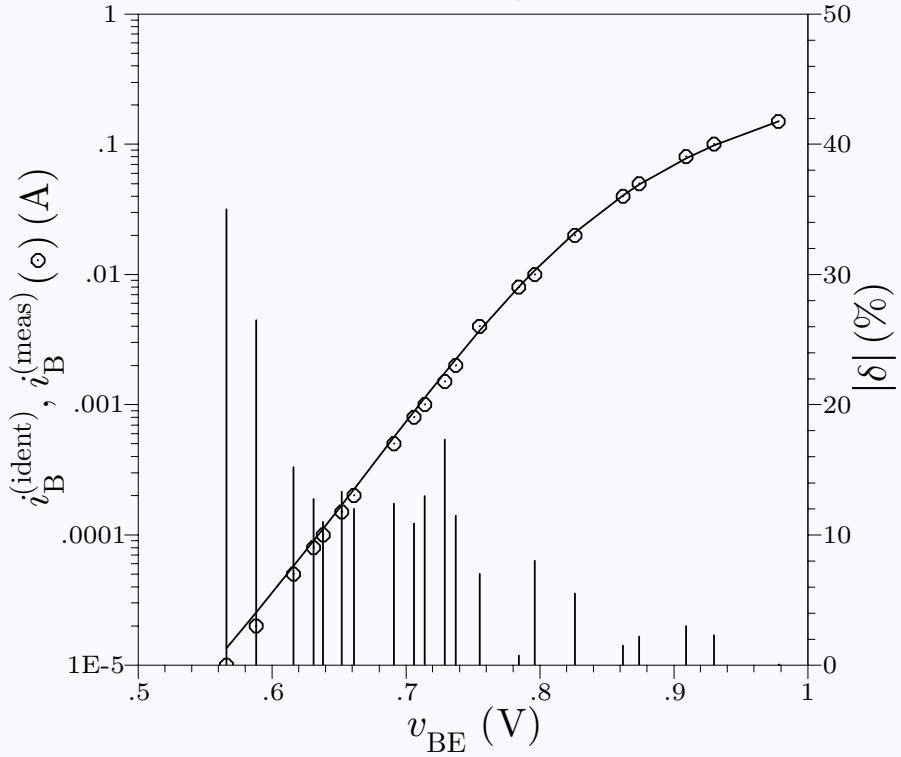
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Forward DC characteristics of BJT KC508 (rms = 3.51 %, δ_{max} = 14.9 % – now with the quasisaturation model part, compare with the first result):



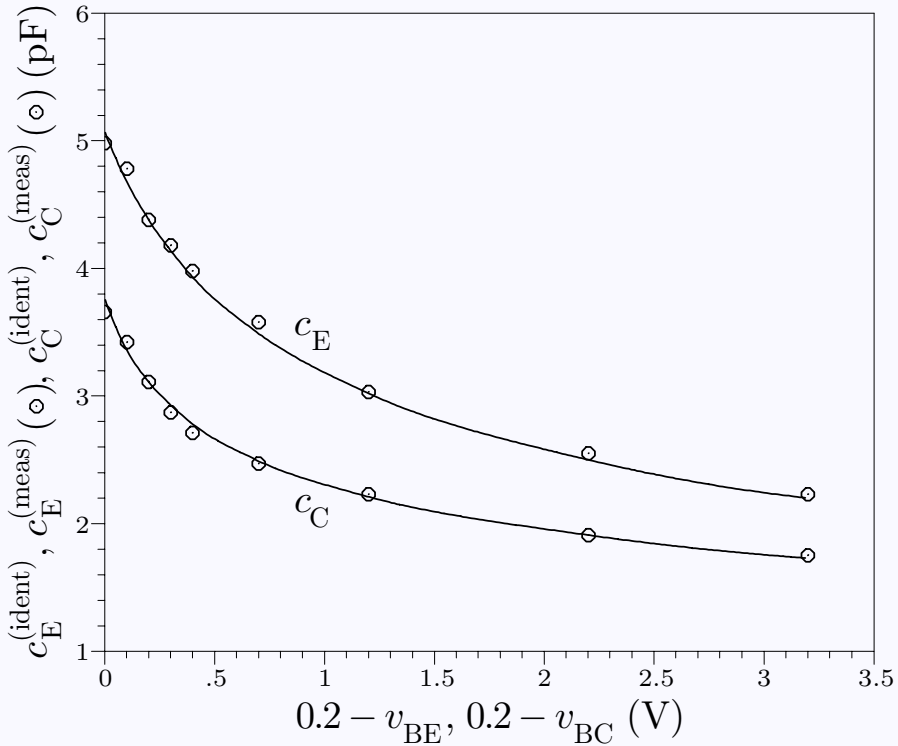
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Input DC characteristic of BJT KC508 with histogram of the errors (with the collector disconnected, rms = 13.5 %, δ_{max} = 35.0 %):



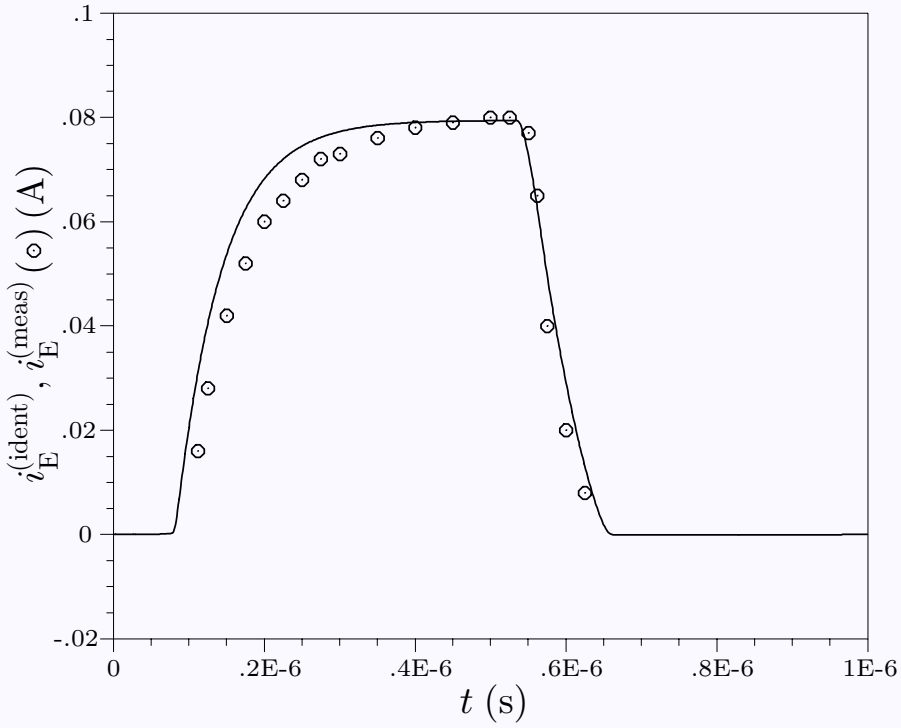
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Collector and emitter junction capacitances of BJT KC508
(rms = 1.57 % (E), 1.64 % (C), and $\delta_{\max}$ = 2.51 % (E), 2.73 % (C)):



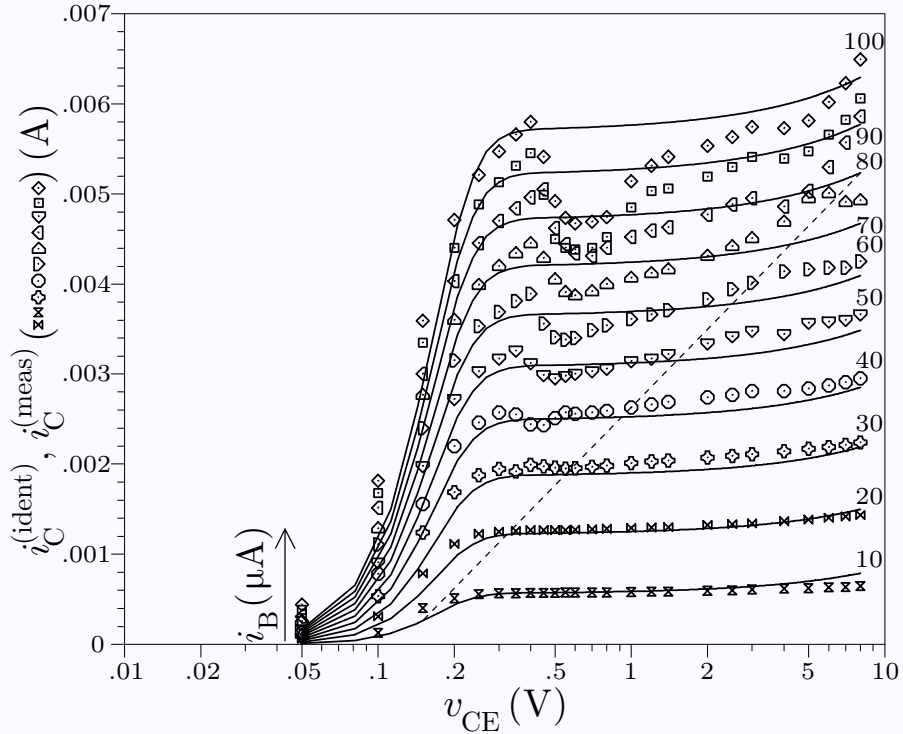
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Identification of transit time model parameters of KC508
(rms = 31.8 %, δ_{max} = 94.4 % – due to the “vertical” error measure):



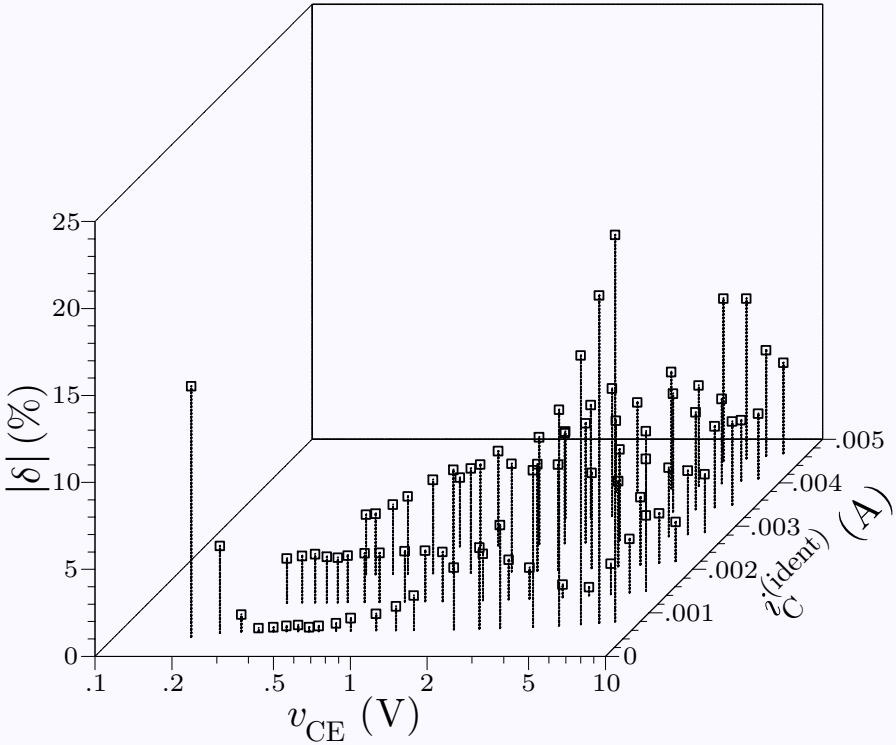
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Forward DC characteristics of microwave BJT KT391 (rms = 16.0 %, δ_{max} = 61.7 % – oscillations occurred during the measurement):



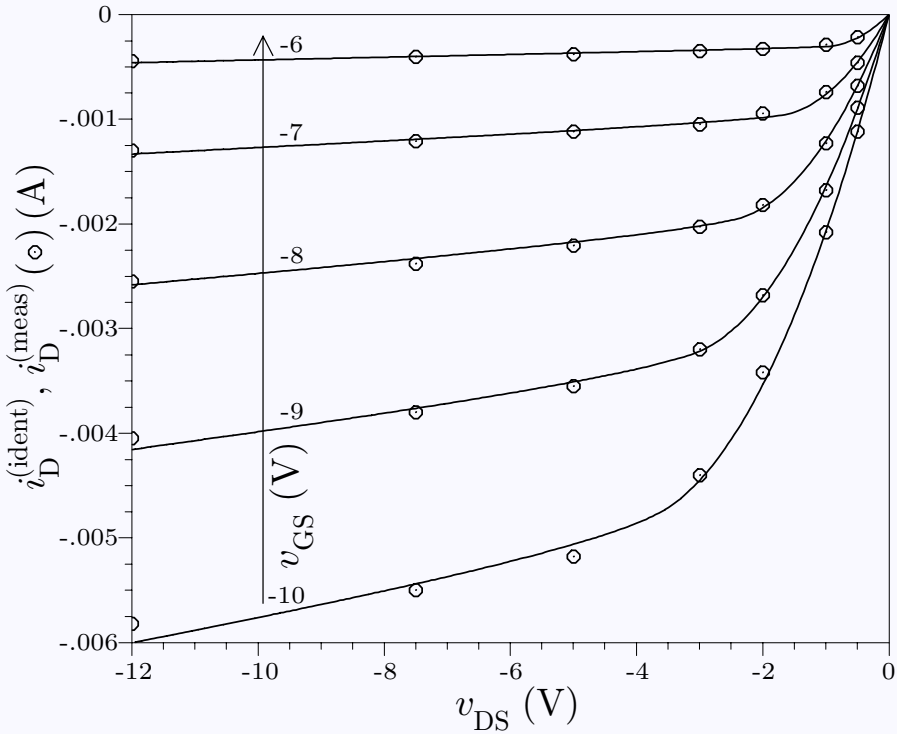
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Relative identification errors in the selected stable area (rms = 5.99 %, δ_{max} = 22.2 % – compare with the previous result):



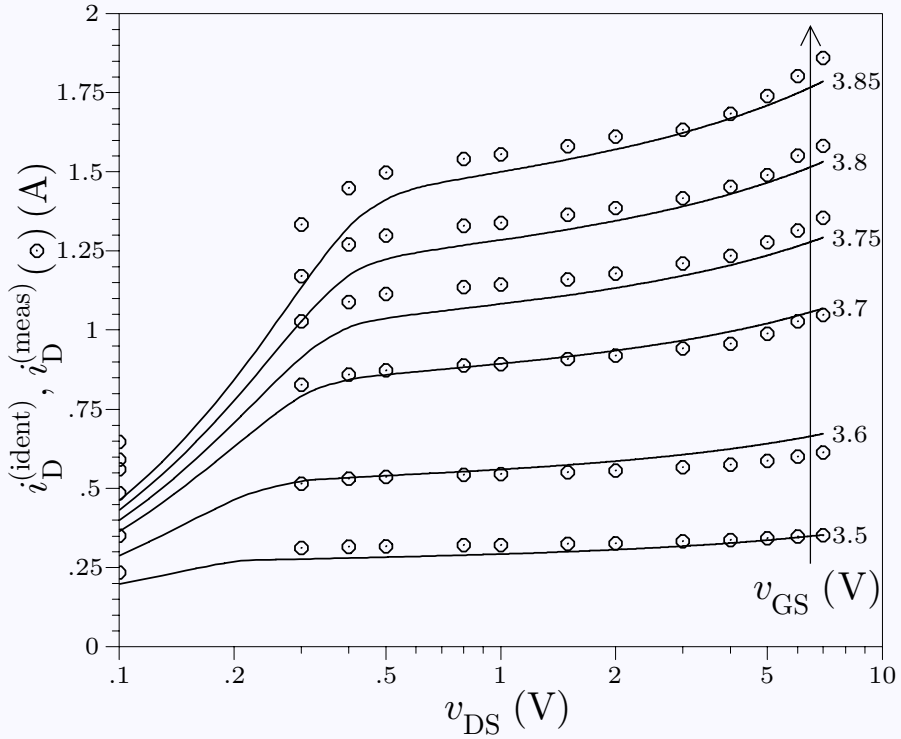
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Forward DC characteristics of (enhancement) PMOSFET 2N3608 (rms = 2.18 %, δ_{max} = 5.41 % only):



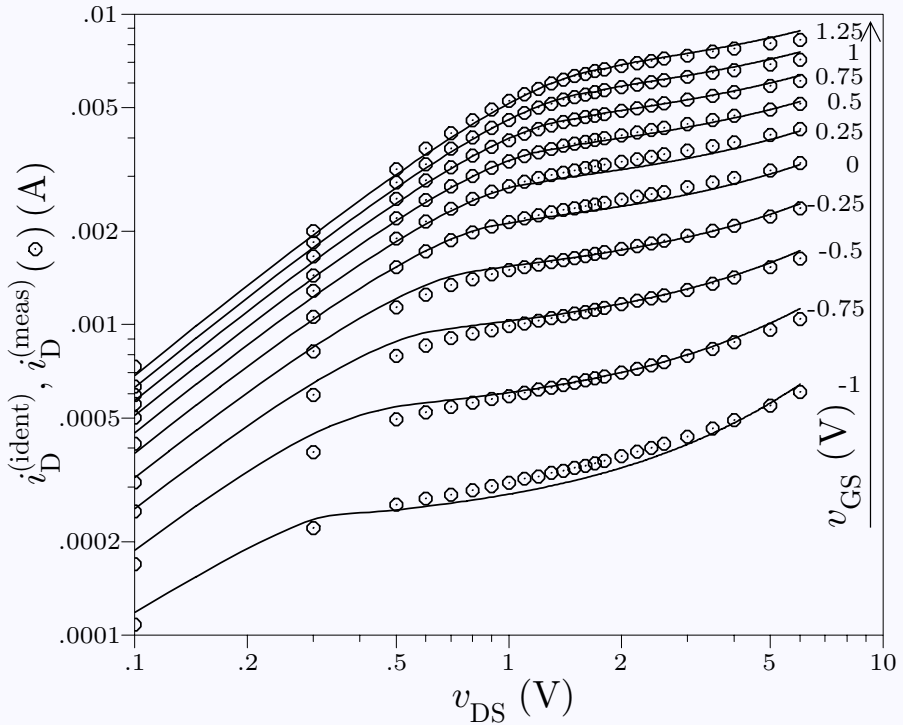
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Forward DC characteristics of (enhancement) NMOSFET BUZ345 (rms = 8.67 %, δ_{max} = 28.8 % – they are greater for power devices):



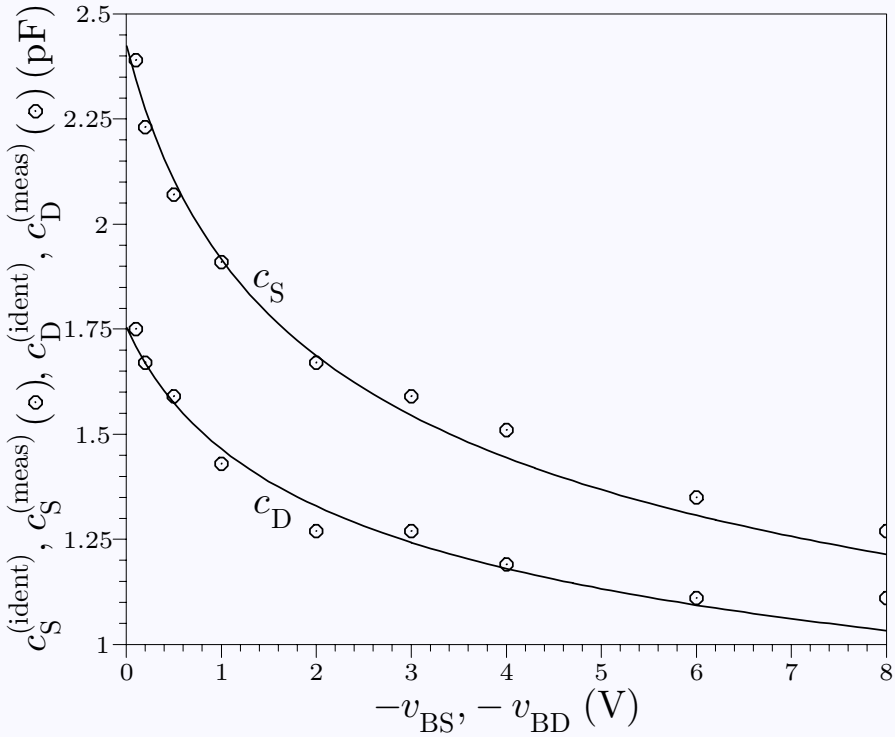
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Forward DC characteristics of (depletion) NMOSFET KF521 (this is a classical Czech device, rms = 4.06 %, $\delta_{\max}$ = 14.5 %):



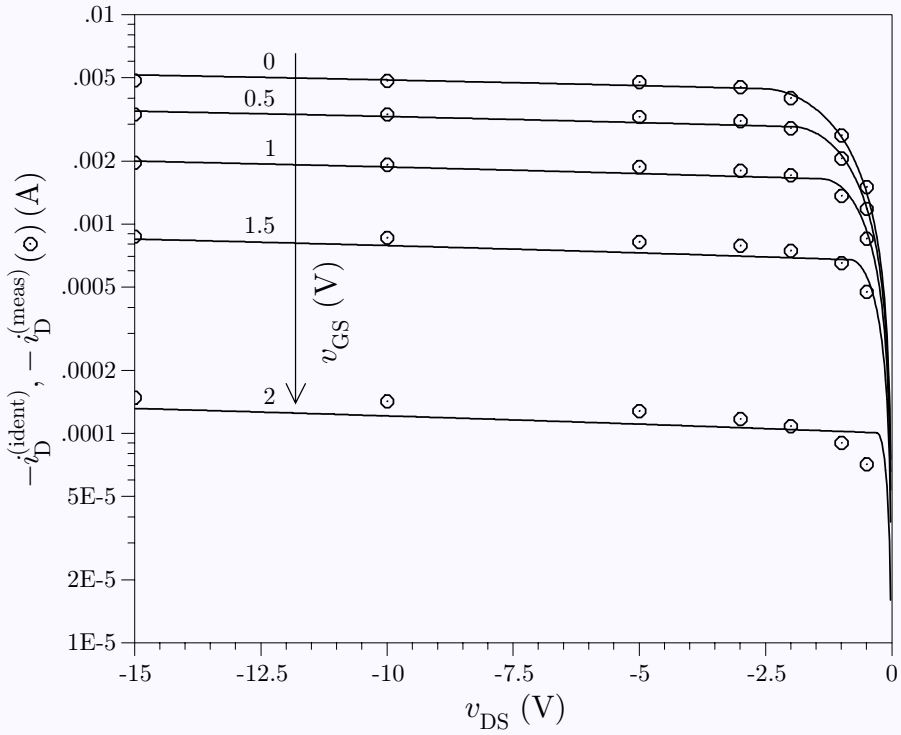
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Drain and source junction capacitances of KF521 (rms = 2.73 % (S), 3.15 % (D), and $\delta_{\max}$ = 4.36 % (S), 6.90 % (D)):



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Forward DC characteristics of PJFET 2N2498 (rms = 10.4 %, $\delta_{\max}$ = 42.38 % – due to the origin):



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Optimization Procedures

- A simple and reliable procedure is used for selecting the scalar factor of the Levenberg-Marquardt method
- The normalization is applied not only on the differences, but also on Jacobian, which is very important for the numerical stability of the task



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- The normalization is applied not only on the differences, but also on Jacobian, which is very important for the numerical stability of the task

Measurement and Identification

- A method for sequential obtaining all the model parameters is verified
- Both LF and HF BJTs have been measured and identified
- Both enhancement- and depletion-mode MOSFETs have been measured and identified
- A typical JFET has been measured and identified



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